

**SECTION A**

General Unit Information:

Unit Name	Undergraduate Project In Software Engineering
Unit Code	CIS050-3
Level	6
Credit Value	30
Location(s) of Delivery	On Campus and Off Campus
Period(s) of Delivery	60 weeks
Pre-requisites or Restrictions	N/A
Aims and Relevance	Whether you go on to further study or into employment one of the most important skills expected in an Honours Graduate is the ability to work autonomously. This unit is an ideal vehicle for the development of that skill. You will typically implement a project proposal developed during the first three weeks of the unit. You will then be asked to contextualize your project idea within current or emerging technologies, typically underpinned by academic publications and to do systematic market research that will equip you with a sound strategy on how to implement your final year undergraduate project. Although you will be working independently, you are expected to arrange regular meetings with your supervisor to guide and support you. You will apply coherently your technical, analytical, practical and managerial skills during your project development. The project is an important opportunity to expand your experience in an area which is of close relevance to the course that you are studying. Your project may be of relevance to a future employer, and will demonstrate your interest and ability in the chosen area. As such your project topic should be aligned with your career aspirations. The following aims are addressed in particular: • To provide you with an opportunity to work autonomously on a sustained piece of work. • To develop your ability to manage a project from start to finish. • To develop your ability to identify problems, analyse the requirements and produce and evaluate solutions. • To develop your ability to work to the relevant academic and professional standards.

Syllabus Content	• Introduction to the final year project • Research Methodology • Choosing a project, • Resources for research – using the academic databases • Critical thinking and critical evaluation of articles • Paraphrasing and plagiarism • Report writing • Software life cycle models • Project Management, WBS and scheduling, GANTT charts • Ethical issues for projects • Data collection from experiments, analysis and presentation • Testing strategies • Identify requirements for artefact as appropriate • Define artefact specification as appropriate • Perform appropriate design activities on artefact • Execute development of physical/conceptual artefact • Perform appropriate analysis, evaluation and testing • Document the artefact and the project activity in a final report
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Learning Outcomes

On completion of this unit you should be able to

1. Demonstrate the following knowledge and understanding

Systematically research the context of a project and underpin the findings with primary data collected using an established research methodology.

2. Demonstrate the following skills and abilities

Manage and complete a major project related to Software Engineering by creating a software artefact using the rigorous application of a software development model (such as waterfall, UML or prototyping) based on customer or client requirements.

PSRB Outcomes and/or Apprenticeship Standard

N/A

Summary Learning Hours

	Scheduled	Guided	Independent	Autonomous	Placement	Total
Hours	24	132	204	240	0	600
Percentage	4%	22%	34%	40%	0%	100%

Approach to Learning

A series of lectures will be used to introduce the main topics. Students will be asked to complete a number of exercises including the identification of sources, provision of a précis of one or more sources, and the correct referencing and citation of those sources. Feedback will be provided by staff to ensure that students are able to reflect further on their work and can improve in future submissions.

During this unit, you will be allocated to a project supervisor who will continue to support you during the major project unit that follows. The supervisor will help you to develop your project proposal and will provide feedback and guidance throughout. You will be expected to take the initiative and arrange these meetings (one meeting per week of approximately 20 minutes).

A centrally managed process will be used to screen your project proposal in the first three weeks of the unit to ensure that it is suitable for your course, adheres to ethical standards, builds upon material you have covered in other units and is appropriately challenging for level three studies.

Students are required to submit project progress reports every four weeks to evidence continuous progress with their project and meaningful interaction with the supervisor.

You will also attend the 'poster exhibition day' where your project is presented to University staff and invited guests. Guidance will be given on how to prepare a relevant poster and you may also use this poster to illustrate your work in the presentation (Assessment 3).

Assessment Summary

No	Assessment Method	Learning Outcome(s)	Weight	Submission Week	Length of Assessment
1	Coursework - Artefact	1 and 2	35	26	one artefact
2	Coursework - Project Report	1 and 2	35	28	10000 words
3	Practical - Oral Presentation	1 and 2	30	30	15 minutes

Learner Development

This Unit particularly focuses on the development of your abilities in the following areas:

Enquiry	You will autonomously develop your own ideas based on your project proposal and previous experience.
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Contextual Understanding, including where employer based	You will contextualize your major project within other units of this course: in particular within the ideas developed as part of your project proposal.
Collaboration	You will work with the unit tutor and with other stakeholders There may also be interaction with your peers for instance if they work on related projects..
Enterprise	You will develop a novel artefact that builds on your previous work.
Assessment 1 Details	You will produce a software engineering project artefact which is the culmination of your project work. The development can be driven by an (external) customer, a current problem or an otherwise identified need.
Assessment 2 Details	You will submit an individual written report of 10,000 words (approximately 70 pages. Excluding appendices) closely relating to the artefact you have produced.
Assessment 3 Details	You will deliver a 15-minute demonstration of your artefact discussing the project that you have undertaken that will use your artefact and an A1 poster.
Assessment 4 Details	

Threshold Expectations

In order to pass Assessment 1 you will need to :

Perform primary research that aligns to secondary research (e.g. academic journals) undertaken.

Implement an artefact and assess it against a given set of criteria.

In order to pass Assessment 2 you will need to:

Use and reference peer reviewed journals and other sources of information relevant to your project in order to arrive at a solution

Clearly articulate the requirements for the artefact

Summarize the design for the artefact

Illustrate the stages of implementation and testing

Effectively communicate your project problem and solution to an audience that consists of both experts and non-experts

Demonstrate proficiency within the subject and ownership of your work through formal academic writing

In order to pass Assessment 3 you will need to:

Effectively communicate your project problem and solution to an audience that consists of both experts and non-experts

Demonstrate proficiency within the subject and ownership of your work through an oral presentation.

In order to pass Assessment 4 you will need to:

Section B

Recommended Reading

Core Text - this unit is supported by the following core text:

Dawson, C. W. (2015a) Projects in computing and information systems [ebook]: a student's guide. Third edition. Harlow: Pearson Education.

Guided Reading

The following is expected reading for this unit. Details of what to read and when will be provide in the BREO site.

McMillan, Kathleen and Weyers, Jonathan D. B. (2011a) How to write dissertations & project reports. 2nd ed. Harlow: Prentice Hall.

Independent Study

To receive high grades you will need to demonstrate your wider reading. The following resources provide useful background reading for the material in this unit. This is not an exhaustive list and students should read widely from the variety of journals available in the Learning Resources Centre.

You should discuss with your supervisor on what best to focus on and which literature best serves the project.

Equality Impact Assessment

Question	Y/N/NA	Additional anticipatory adjustments/actions if necessary
Learning materials will be made available in advance of sessions for students to adapt as appropriate?	Y	
The approach to teaching and learning is sufficiently flexible to enable all students to succeed?	Y	
The approach to group work takes account of the needs of students with disabilities and from diverse backgrounds?	Y	
The approach to practical work takes account of the needs of students with disabilities?	Y	
Students with a protected characteristic* have an equal opportunity to achieve the learning outcomes?	Y	

The assessment tasks provide all students with an equal opportunity to succeed?	Y	
Any other aspects of the unit that might pose potential challenges from an equality or diversity perspective have been considered?	NA	

* Age, Gender reassignment, Marriage and civil partnership, Pregnancy and maternity, Race, Religion and belief, Sex, Sexual orientation

Section C

Administrative Information – Faculty completion	
Faculty	Creative Arts Technologies and Science
Portfolio	CU-COM
School/Department	School of Computer Science and Technology
HECOS Code(s)	100374
Unit Co-ordinator	Enjie Liu
Apprenticeship ?	No

Courses that this unit is delivered on
BSc (Hons) Software Engineering